Practice workbook · Explain diffusion in a context and factors affecting its rate

Diffusion of gases or solutes in cells, bacteria, gut and dialysis, and how surface area to volume ratio and concentration gradient change rate.

7 question parts: 1 worked-example retry and 6 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: HBI-EXPLAIN-DIFFUSION-RATE-IN-CONTEXT

1. Question 5 · part a · 3 marks

June 2025 Paper 2R

Worked-example retry · Attempt part a only.

5 Glucose is a product of digestion.

Glucose moves from the small intestine into the blood by diffusion.

(a) Explain why the rate of diffusion of glucose into the blood stream is affected by the concentration of glucose in the small intestine.

(3)

(b) A student uses different concentrations of glucose solution to investigate the rate of diffusion from the small intestine into the blood.

The list gives some of the equipment that the student uses in the investigation.

- Visking tubing (to represent the small intestine)
- boiling tube
- different concentrations of glucose solutions
- stop clock

(i) Name the chemical the student uses to test for glucose.

(1)

(ii) Name one other piece of equipment that the student would need to use to test for glucose.

(1)

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1. Question 5 · part a · 3 marks · Working and answer

Working and answer

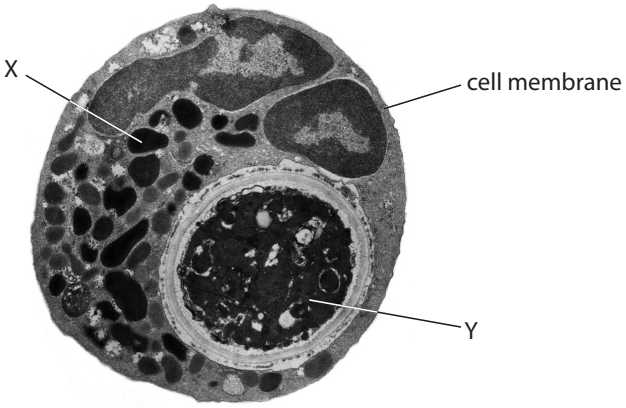
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2. Question 2 · part b · 2 marks

June 2019 Paper 2

Further practice · Attempt part b only.

2 (a) The image is an electron micrograph of a human body cell.



(Source: © Science History Images/Alamy)

(i) The table gives information about parts of the human body cell.

Complete the table by giving the missing information.

(2)

Part	Name of part	Function
X		release energy from glucose
Y	nucleus	

(ii) Electron microscopes and light microscopes can both be used to view body cells.

Which of these is an advantage of using an electron microscope?

(1)

- ☒ **A** more cell structures can be seen with greater resolution
- ☒ **B** more cell structures can be seen with less resolution
- ☒ **C** fewer cell structures can be seen with greater resolution
- ☒ **D** fewer cell structures can be seen with less resolution

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2. Question 2 · part b · 2 marks

June 2019 Paper 2

Further practice · Attempt part b only.

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(iii) The actual diameter of this human body cell before magnification is 0.05 mm.

Use information from the image to calculate the magnification of the cell.

(3)

magnification = ...

(b) Body cells produce carbon dioxide.

Describe how carbon dioxide passes from a body cell into the blood.

(2)

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(Total for Question 2 = 8 marks)

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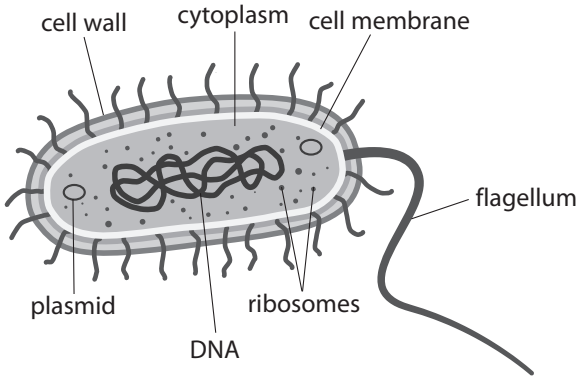
Working and answer

3. Question 3 · part b.i · 1 mark

January 2020 Paper 1

Further practice · Attempt part b.i only.

3 (a) The diagram shows a type of bacterial cell.



(i) The boxes give some structures found in the bacterial cell and some functions of cell structures.

Draw one straight line from each structure to its correct function.

(3)

Structure	Function
cell membrane	protects against pathogens
	helps the cell to move
ribosomes	makes proteins
	holds the genetic code
flagellum	
	controls which substances enter and exit the cell

(ii) State two structural differences between the bacterial cell and a sperm cell.

(2)

1 .....

.....

2 .....

.....

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3. Question 3 · part b.i · 1 mark

January 2020 Paper 1

Further practice · Attempt part b.i only.

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(b) Many bacteria absorb oxygen and nutrients from their environment. Oxygen and nutrients are used by bacteria in aerobic respiration.

(i) Which process transports oxygen into bacterial cells?

(1)

- ☒ **A** active transport
- ☒ **B** diffusion
- ☒ **C** excretion
- ☒ **D** osmosis

(ii) Complete the word equation for aerobic respiration.

(2)

oxygen + ... → ..... + water

(iii) Explain why respiration is important for bacteria.

(2)

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(Total for Question 3 = 10 marks)

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Working and answer

4. Question 6 · part c · 2 marks

June 2024 Paper 1

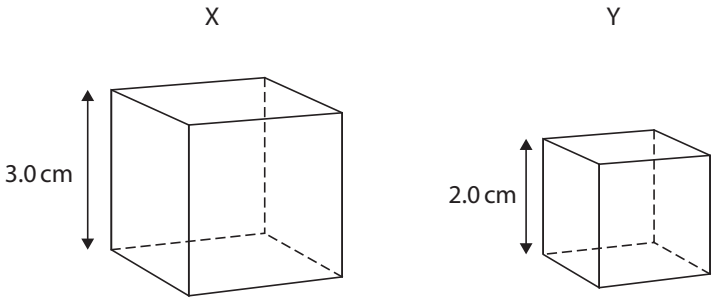
Further practice · Attempt part c only.

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6 (a) Humans have a small surface area to volume ratio (SA : V) compared with some other organisms.

Cube X represents a human.

Cube Y represents an organism with a larger SA : V.



The table gives the SA : V of cube Y.

Cube	Surface area in cm^2	Volume in cm^3	SA : V
X			
Y	24.0	8.0	3:1

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(i) Calculate the surface area of cube X.

(2)

surface area = ... cm^2

(ii) Calculate the volume of cube X.

(2)

volume = ... cm^3

(iii) Calculate the SA : V of cube X.

(2)

SA : V = ... : 1

4. Question 6 · part c · 2 marks

June 2024 Paper 1

Further practice · Attempt part c only.

(b) Multicellular organisms, with a small SA:V ratio, such as humans, need an efficient transport system to deliver substances to body cells.

Describe the role of the human blood transport system in delivering substances to body cells.

(5)

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4. Question 6 · part c · 2 marks

June 2024 Paper 1

Further practice · Attempt part c only.

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(c) Bacteria are able to exchange materials directly with their environment as they have a very high SA : V ratio.

They do not have an internal transport system.

Describe how bacteria exchange materials with their environment.

(2)

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(Total for Question 6 = 13 marks)

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Working and answer

5. Question 3 · part a.iv · 2 marks

January 2023 Paper 2

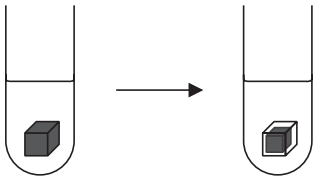
Further practice · Attempt part a.iv only.

3 A student investigates diffusion using agar cubes and an acid.

The agar cube contains an alkali and a pH indicator.

The indicator is pink in alkali and colourless in acid.

Acid diffuses into the cube, causing the cube to change from pink to colourless from the edge.



agar cube when first placed in acid

agar cube starts to become colourless

(a) The cube the student placed in the acid had a side length of 2 mm.

(i) Calculate the surface area to volume ratio of the cube.

Your answer should be in the form n : 1

(3)

surface area to volume ratio = ...

(ii) The cube took 1 minute and 32 seconds to become colourless.

Which of these shows the time in seconds?

(1)

- ☒ **A** 62
- ☒ **B** 82
- ☒ **C** 92
- ☒ **D** 102

(iii) State what is meant by the term **diffusion**.

(1)

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5. Question 3 · part a.iv · 2 marks

January 2023 Paper 2

Further practice · Attempt part a.iv only.

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(iv) Explain the effect of increasing the surface area to volume ratio on the time taken for the cube to become colourless.

(2)

(b) A student wants to investigate the effect of temperature on the rate of diffusion into the agar cube.

(i) Describe a method for this investigation using the apparatus shown in the diagram.

(2)

(ii) Give two variables the student would need to control in their investigation.

(2)

1 .....

2 .....

(Total for Question 3 = 11 marks)

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5. Question 3 · part a.iv · 2 marks · Working and answer

Working and answer

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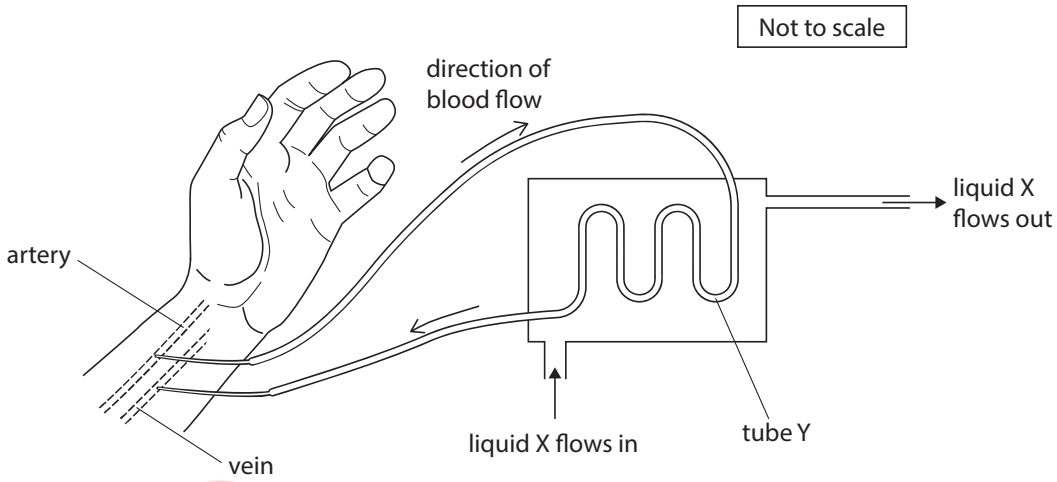
6. Question 5 · part a.iii · 2 marks

June 2025 Paper 1R

Further practice · Attempt part a.iii only.

5 The diagram shows a kidney dialysis machine.

The wall of tube Y is partially permeable so it allows the passage of small molecules.



(a) (i) Explain why the blood flow is in the direction shown in the diagram.

(4)

(ii) State which part of the kidney nephron has the same role as tube Y.

(1)

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P 7 4 6 2 8 A 0 1 2 2 0

6. Question 5 · part a.iii · 2 marks

June 2025 Paper 1R

Further practice · Attempt part a.iii only.

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(iii) Explain why tube Y has a length of several metres in the dialysis machine.

(2)

(iv) State one difference between the composition of liquid X as it flows into the machine and as it flows out of the machine.

(1)

(b) Glucose is added to liquid X before it passes into the dialysis machine.

Explain why glucose is added.

(3)

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6. Question 5 · part a.iii · 2 marks · Working and answer

Working and answer

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7. Question 7 · part a.ii · 2 marks

June 2022 Paper 2

Further practice · Attempt part a.ii only.

7 Read the passage below.

Use the information in the passage and your own knowledge to answer the questions that follow.

Our body is under constant attack from free radicals. Free radicals are molecules produced during chemical reactions that take place in body cells. Free radicals can change the structure and function of cells and damage genetic material. Some scientific evidence also suggests that free radicals can affect the flow of substances that enter and leave a cell.

Vitamin C is water-soluble and an important component of the daily diet. The recommended intake of vitamin C is 90 mg per day. Vitamin C helps to strengthen the immune system and improves the absorption of iron into the body. Vitamin C is also known as an antioxidant. Antioxidants protect cells from damage by free radicals and other substances by changing the way that free radicals behave.

Water and high temperatures can reduce the vitamin C content of food. For example, 40 g of fresh broccoli provides 80% of the recommended daily intake of vitamin C. The same mass of boiled broccoli provides 60% of the recommended daily intake of vitamin C. Boiling is the biggest cause of vitamin C loss, although baking and frying also lower vitamin C content.

(a) (i) Free radicals can damage genetic material in cells.

Describe one effect that this could have on body cells. (lines 2 and 3)

(2)

(ii) Suggest how antioxidants, such as vitamin C, allow cells to function normally. (lines 5 and 6 and lines 8 to 10)

(2)

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7. Question 7 · part a.ii · 2 marks

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Further practice · Attempt part a.ii only.

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(iii) Calculate the mass, in grams, of boiled broccoli that should be eaten to provide the recommended daily intake of vitamin C. (lines 12 to 14)

(3)

mass = ... g

(b) DCPIP is a chemical used to determine the amount of vitamin C in food.

Describe a method, using DCPIP, to compare the amount of vitamin C in fresh orange juice with that in boiled orange juice.

(5)

(Total for Question 7 = 12 marks)

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7. Question 7 · part a.ii · 2 marks · Working and answer

Working and answer

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